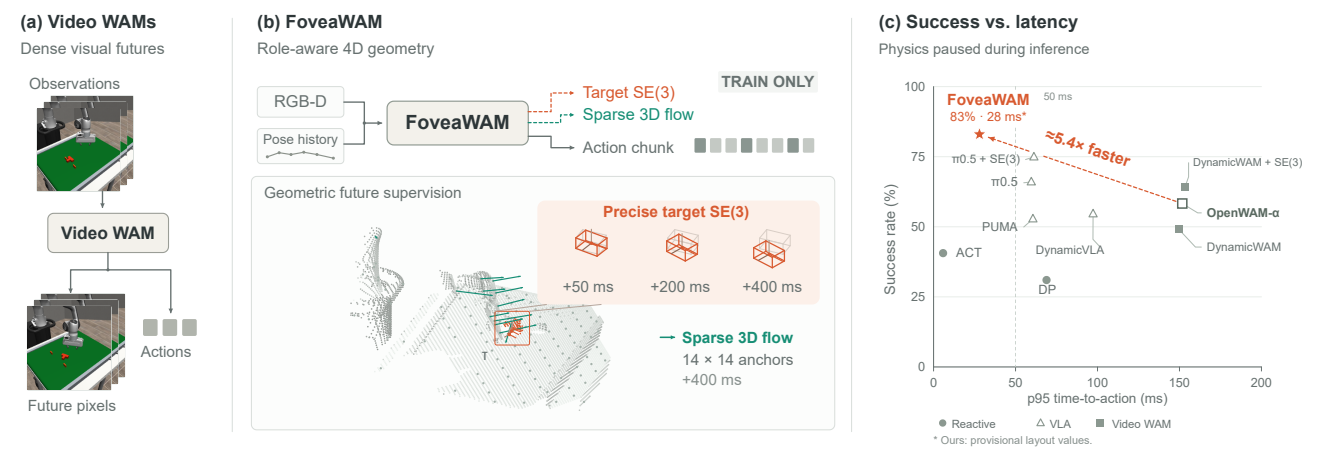

(a) Video WAMs
Dense visual futures
Observations
Video WAM
Actions
Future pixels
(b) FoveaWAM
Role-aware 4D geometry
RGB-D
Pose history
FoveaWAM
Target SE(3)
Sparse 3D flow
Action chunk
TRAIN ONLY
Geometric future supervision
Precise target SE(3)
+50 ms
+200 ms
+400 ms
Sparse 3D flow
14 × 14 anchors
+400 ms
(c) Success vs. latency
Physics paused during inference
FoveaWAM
83% · 28 ms*
50 ms
≈5.4× faster
π0.5 + SE(3)
π0.5
PUMA
ACT
DP
DynamicVLA
DynamicWAM + SE(3)
OpenWAM-α
DynamicWAM
Success rate (%)
p95 time-to-action (ms)
Reactive
VLA
Video WAM
* Ours: provisional layout values.